SAVEETHA
INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Programme: **B.E / B.Tech** **Name : L.Deepthi (192411213)**

Course Code: **CSA6303**

Course Name: **Threat Intelligence and Network Security**

Duration: **1: 30 Hr** ·

Date : Max Marks:
50 Marks

CONTINUOUS INTERNAL ASSESSMENT 1

Course Outcomes

On successful completion of the course, students will be able to:

CO	Course Outcome Statement
CO1	Interpret security principles, network threats, and the role of threat intelligence in cybersecurity.
CO2	Identify common network attacks and analyze threat intelligence sources and data collection techniques.

Answer all the Questions

1. Medilife Hospital runs a patient-records portal that doctors, nurses and insurers access over the hospital network. In one week the security team logs three incidents: (i) ransomware encrypts the radiology database and staff cannot open scans, (ii) a disgruntled employee silently changes a patient's blood-group entry in the lab system, and (iii) an attacker captures unencrypted patient data travelling on the Wi-Fi network without altering it. Interpret each incident against the CIA triad, identifying the security goal that is primarily violated and justifying your choice. Classify each incident as a passive or active threat and as an internal or external threat. Recommend one preventive control for each incident, explaining how it restores the violated goal. Then analyse how the three goals can conflict with one another when the hospital designs access to emergency

patient records, and explain why a balance among confidentiality, integrity and availability is required for a hospital network.

(10) (CO1) (BL3)

2. The Security Operations Centre (SOC) of a national bank receives four intelligence items in a single morning: (i) a quarterly report from a consulting firm stating that financially motivated groups are expected to target mobile-banking customers over the next year, (ii) a list of malicious IP addresses and domain names linked to a phishing campaign, (iii) an advisory describing the attacker's planned timing, tools and likely targets for an imminent attack on the bank's payment gateway, and (iv) a write-up of the attacker's tactics, techniques and procedures (TTPs) in a recent banking-malware campaign. Classify each item as strategic, tactical or operational threat intelligence, justifying your choice. Compare the three types of threat intelligence in terms of
(a) target audience and decision level,
(b) time horizon and level of technical detail, and
(c) typical sources and the action taken by the bank.

   Finally, analyse what the bank would lose if it relied only on tactical intelligence (IoC lists) and ignored strategic and operational intelligence.

   (10) (CO1) (BL4)

3. a) An e-commerce company wants to build a threat intelligence programme after a data breach. Apply the threat intelligence lifecycle to the company by describing, for each phase (direction/planning, collection, processing, analysis, dissemination and feedback), the activities carried out and the output produced. Explain how the feedback phase keeps the lifecycle continuous, and give two applications of threat intelligence in the IT industry (such as SOC operations, incident response or vulnerability prioritisation) that the company can adopt.

   (5) (CO1) (BL4)

   b) An employee in the finance department receives an email that appears to come from the CEO, urging an immediate payment and containing a link to a "secure invoice portal". The employee clicks the link, enters credentials and downloads an attachment, after which files on the shared drive become inaccessible. Analyse the incident by identifying the types of attack involved (phishing and malware), the stages of the attack from email delivery to impact, and the indicators in the email that should have raised suspicion. Suggest two technical and two user-awareness measures to prevent a recurrence.

   (5) (CO2) (BL4)

4. A university campus network experiences three problems: (i) the online examination portal becomes unreachable when thousands of requests per second arrive from many compromised devices around the world, (ii) students connected to the free campus Wi-Fi are redirected to a fake login page, and their traffic is relayed through an attacker's laptop that reads and modifies the data, and (iii) an attacker sends packets with a forged source address of the registrar's server to gain access to the administrative VLAN, while another user runs a packet-capture tool to read unencrypted traffic. Identify the attack in each case and explain how it works. Distinguish between DoS and DDoS attacks, between spoofing and sniffing, and between sniffing (passive) and a man-in-the-middle

attack (active). For each attack, describe one detection technique and two countermeasures (for example, rate limiting and traffic scrubbing, HTTPS/TLS with certificate validation, ARP inspection, encryption and ingress filtering) that the university should deploy. Also explain how a vulnerability assessment of the campus network would help reduce the likelihood of these attacks.

(10) (CO2) (BL4)

5. A threat-intelligence analyst at a managed security provider receives a commercial threat feed containing the following entries related to a ransomware campaign: IP address 203.0.113.45, domain "secure-login-update[.]com", file hash (SHA-256) of a malicious dropper, URL "hxxp://secure-login-update[.]com/invoice.exe", and the email subject line "Urgent: Invoice Overdue". The provider also wants to supplement the feed with open-source intelligence (OSINT). Analyse the feed by classifying each entry as a network, host or email indicator of compromise (IoC) and evaluating which entries are easy or hard for an attacker to change. Explain how the analyst should assess the quality of the feed (timeliness, accuracy, relevance and context) before deploying the IoCs to firewalls and SIEM tools, and the risk of false positives. Identify four OSINT sources and the data collection techniques (such as monitoring social media, paste sites, public breach databases, domain WHOIS and passive DNS) that can enrich the feed, and compare OSINT with commercial threat feeds in terms of cost, reliability and coverage. Finally, justify which combination of sources the provider should use.

(10) (CO2) (BL4)

CSA6303 Continous Internal Assessment - I

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Threat Intelligence & Network Security

(1) Introduction:- The CIA Traid is a fundamental model of Information Security. It consists of three important security goals:

1. Confidentiality
2. Integrity
3. Availability

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Confidentiality	Integrity	Availability.
Ensures that sensitive Information is accessed only by authorized people.	Ensures that Information remains accurate and is not modified without permission.	Ensures that Information and services are accessible whenever required.

Analysis of the three security incidents:

1. Ransomware encrypts the radiology database.

* Security goals voiolated: Availability.
* Reason: Doctors and staff cannot access radiology reports and patient records because the database is encrypted.

* Attack type:- Active attack
* Threat source:- External threat.
* Classification: Malicious attack.
* preventive control:-
    * Maintain regular offline offline backups.
    * Install updated antivirus and endpoint detection software.
    * Use network segmentation.

CIA Triad in Hospital Security.

Confidentiality.
protect patient Information

Integrity
Accurate medical records

Availability
Accessible hospital systems.

2. Disgruntled Employee changes a patient's blood group Entry.

* Security goals violated : Integrity.
* Reason: The employee changes medical Information without authorization, making the patient's record incorrect.

3. Attacker captures unencrypted patient data Travelling over Wi-Fi

* Security goal violated : Confidentiality.
* Reason :- The attacker reads sensitive patient information during network transmission.
* Attack type : passive attack.
* Threat source : External threat.

2. Threat Intelligence in a National Bank.

Introduction :- Threat Intelligence is the process of collecting, analysing and sharing information about cyber threats to help organizations identify and prevent attacks.

The three major types of threat Intelligence are :

1. Strategic Intelligence.
2. Tactical Intelligence
3. Operational Intelligence.

Classification of the four Intelligence items :-

Intelligence item	Type	Reasons.
Quarterly report about	Strategic	provides long-terms

about financially motivated groups targeting mobile banking customers	Strategic	provides long-term threat trends & Business risks.
List of malicious IP addresses and domain names	Tactical	Provides Indicators of compromise (IOCs) for detecting threats.
Advisory describing attackers planned timing, tools and targets	Operational	Provides details about an imminent attack.
Write-up of attacker's TTPs in banking malware campaign	Tactical	Explains attacker techniques, tactics and procedures.

1. Strategic Threat Intelligence:

→ provides high-level information about cyber threats.

→ Helps management understand long-term risks

→ Focuses on business decisions and security planning.

Tactical Threat Intelligence

Focuses on attacker tactics, techniques and procedures (TTPs)

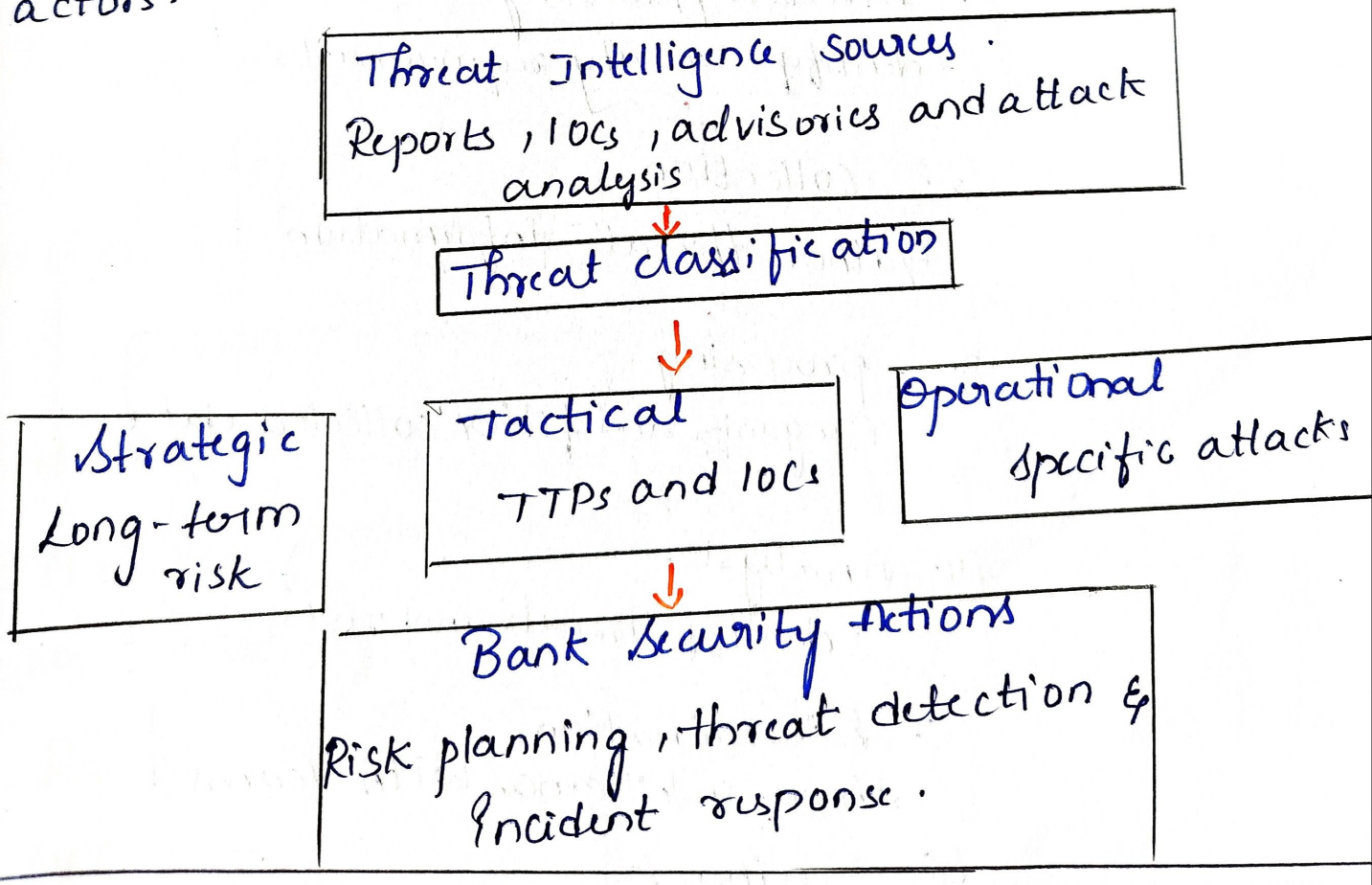
* Helps security teams understand how attackers operate

* Includes IP addresses, malicious domains, file hashes and attack patterns.

3. Operational threat Intelligence.

* provides information about specific planned or ongoing attacks.

* Focuses on attack timing, targets, methods & threat actors.



1. It may fail to understand long-term cyber risks.
2. It may not prepare for future attacks.
3. It may not allocate sufficient security resources.

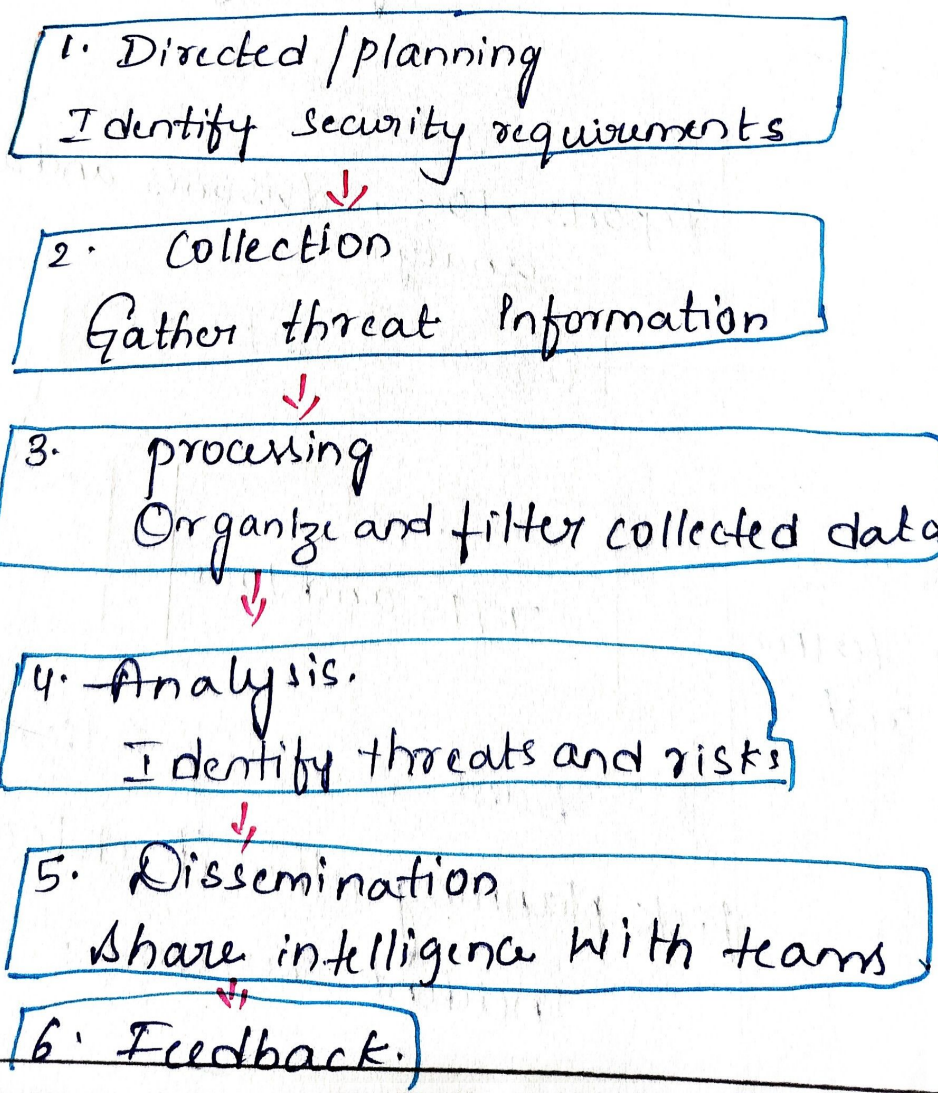
Conclusion: A bank must combine strategic, tactical and operational Intelligence. It helps management plan security.

Q3 a. Threat Intelligence Lifecycle In a E-Commerce Company.

Introduction: The Threat Intelligence Lifecycle is a Continous process used to collect, analyse and use cyber threat information to protect an organization.

Stages of Threat Intelligence Lifecycle:

Flowchart:



...ications of Threat Intelligence in IT:

... operations: Helps security teams monitor suspicious ...tivities and detect cyberattacks.

Incident Response: Helps identify attack quickly and take corrective.

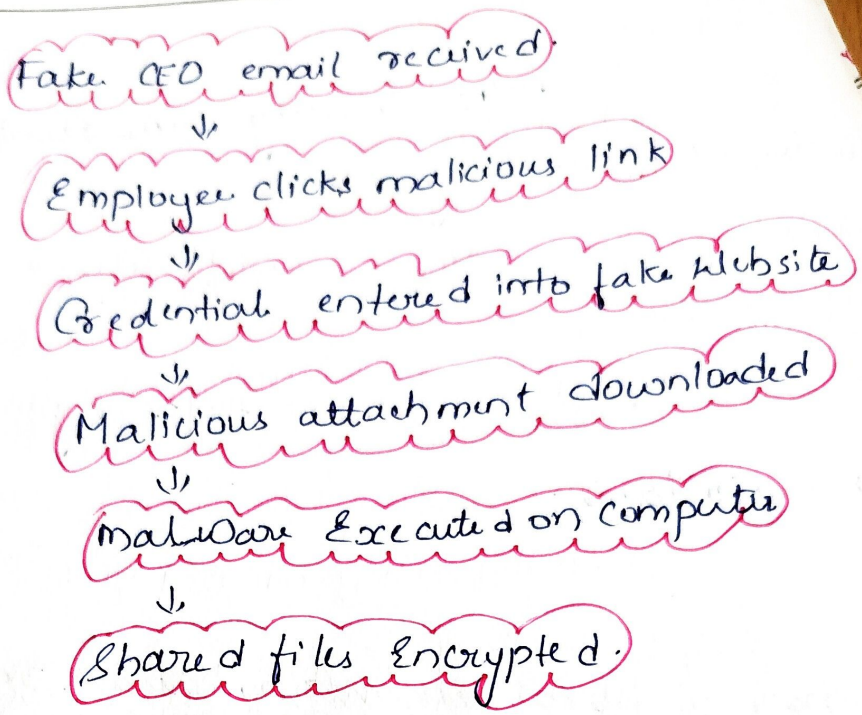
3. Vulnerability prioritization: Helps organization fix highly exploitable.

3b. Introduction:-

Phishing is a social engineering attack in which attackers send fake emails to trick users into revealing sensitive information.

Types of Attacks Involved:-

1. phishing Attack:- The attacker sends a fake email pretending to be the CEO and requests an urgent payment
2. Credential Theft:- The employee enters login details into a fake Invoice portal.
3. Malware Attack:- A malicious attachment is downloaded and executed.
4. Ransomware Attack:- Files on the shared drive become inaccessible because they are encrypted.



49. Introduction:- A university campus network connects students, teachers, administrators and online examination systems.

Attackers may exploit network vulnerabilities to interrupt services, steal information and gain unauthorized access.

1. Distributed Denial of service :- (DDOS)

Meaning:- A DDOS attack occures when multiple compromised devices send a huge number of requests to a target servers, making the service unaviable

ng:-

Hackers control many compromised devices called botnet.

* These devices send thousands of requests to the examination portal.
* The server becomes overloaded.
* Genuine students cannot access the portal.

Detection Technique:-

* Monitor unusual traffic spikes.
* Use network traffic analysis and IDS/IPS tools.

Two Countermeasures:-

1. Implement rate limiting and traffic filtering.
2. Use DDoS protection services and load balancing.

2. Man-In-the-Middle (MITM) Attack:

Meaning:- A man-in-middle attack occurs when an attacker secretly positions themselves between two communicating parties to intercept and possi modify their communication.

Working:-

Working:

→ Students connect to free campus wifi.

→ The attacker creates a fake login page.

→ Students enter their usernames and passwords

→ The attacker may read or modify transmitted information.

Conclusion:- The university should implement firewalls, IDS/IPS encryption, secure authentication, network monitoring and regular vulnerability assessments to protect its campus network from cyber attacks

5): Threat Intelligence Feed Analysis & OSINT.

Introduction:- Threat Intelligence feeds provide information about malicious activities, attackers and Indicators for compromise (IOCs). They help security analysts identify and prevent cyber attacks.

An indicator of compromise (IOC) is a piece of information that indicates a possible security breach.

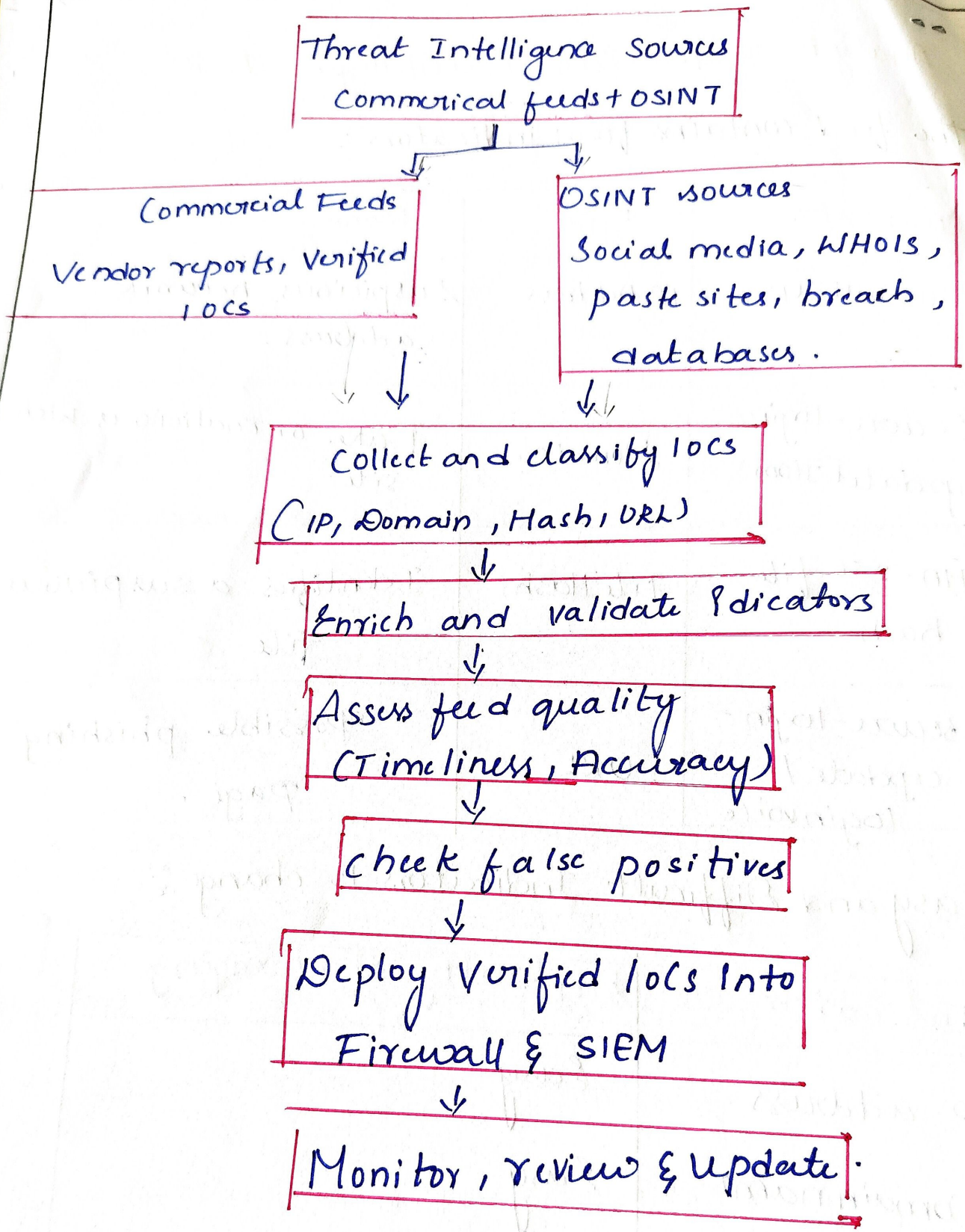
lysis of Given Threat Intelligence Feed:-

feed contains four indicators:

Indicator	type	Meaning.
203.0.113.45	IP Address	Suspicious network address.
vsecure-login update[.]com	Domain	Fake or malicious Web-site.
SHA-256 file hash	File Hash	Identifies a suspicious file
/secure-login-update/loginvoice	URL	Possible phishing Page.

Easy and Difficult Indicators to change:-

Indicator	Difficulty of changing
IP address	Easy
Domain name	Easy
File hash	Easy if file is modified
URL.	Easy


Threat Intelligence Sources
Commercial feeds + OSINT
Commercial Feeds
Vendor reports, Verified IOCs
OSINT sources
Social media, WHOIS, paste sites, breach databases.
Collect and classify IOCs
(IP, Domain, Hash, URL)
Enrich and validate Indicators
Assess feed quality
(Timeliness, Accuracy)
Check false positives
Deploy Verified IOCs into Firewall & SIEM
Monitor, review & update.